GATE CS — Algorithms

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
Graph algorithms: MST, shortest path, BFS/DFS	31	56	2
Asymptotics & recurrences	17	23	2
Sorting, searching, divide & conquer	11	14	2
Hashing	6	7	2
Dynamic programming	3	5	3
Greedy	3	5	2
Total	**71**	**110**	**13**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

Graph algorithms: MST, shortest path, BFS/DFS

MST properties and counts, DFS/BFS, shortest paths. · 31 questions, 56 marks over 15 papers (≈3.7 marks/paper)

A. Past-paper questions (31)

2026 CS1 · Q.41 · 2 marks

Let G(V, E) be an undirected, edge-weighted graph with integer weights. The weight of a path is the sum of the weights of the edges in that path. The length of a path is the number of edges in that path. Let s ∈ V be a vertex in G. For every u ∈ V and for every k ≥ 0, let dk (u) denote the weight of a shortest path (in terms of weight) from s to u of length at most k. If there is no path from s to u of length at most k, then dk (u) = ∞. Consider the statements: S1: For every k ≥ 0 and u ∈ V, dk+1 (u) ≤ dk (u). S2: For every (u, v) ∈ E, if (u, v) is part of a shortest path (in terms of weight) from s to v, then for every k ≥ 0, dk (u) ≤ dk (v). Which one of the following options is correct?

(A) Only S1 is true

(B) Only S2 is true

(C) Both S1 and S2 are true

(D) Neither S1 nor S2 is true

2026 CS1 · Q.49 · 2 marks

Let G(V, E) be a simple, undirected, edge-weighted graph with unique edge weights. Which of the following statements about the minimum spanning trees (MST) of G is/are true?

(A) In every cycle C of G, the edge with the largest weight in C is not in any MST

(B) In every cycle C of G, the edge with the smallest weight in C is in every MST

(C) For every vertex v ∈ V, the edge with the largest weight incident on v is not in any MST

(D) For every vertex v ∈ V, the edge with the smallest weight incident on v is in every MST

2026 CS1 · Q.50 · 2 marks

Consider the following pseudocode for depth-first search (DFS) algorithm which takes a directed graph G(V, E) as input, where d[v] and f[v] are the discovery time and finishing time, respectively, of the vertex v ∈ V. DFS(G): Explore(G, v, t): unmark all v ∈ V mark v t ←0 t ←t+1 for each v ∈ V d[v] ← t if v is unmarked for each (v, w) ∈ E t ← Explore(G, v, t) if w is unmarked end if t ← Explore(G, w, t) end for end if end for t ←t + 1 f[v] ← t return t Suppose that the input directed graph G(V, E) is a directed acyclic graph (DAG). For an edge (u, v) ∈ E, which of the following options will NEVER be correct?

(A) d[u] < d[v] < f[v] < f[u]

(B) d[v] < d[u] < f[u] < f[v]

(C) d[v] < f[v] < d[u] < f[u]

(D) d[u] < d[v] < f[u] < f[v]

2026 CS2 · Q.37 · 2 marks

Let G be a weighted directed acyclic graph with m edges and n vertices. Given G and a source vertex s in G, which one of the following options gives the worst case time complexity of the fastest algorithm to find the lengths of shortest paths from s to all vertices that are reachable from s in G?

(A) $\Theta(m + n)$

(B) $\Theta(m + n \log(n))$

(C) $\Theta(nm)$

(D) $\Theta(n3)$

2025 CS1 · Q.18 · 1 mark

Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let d1 (u, v) and d2 (u, v) be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v?

(A) $d1(u, v) = d2(u, v)$

(B) $d1(u, v) \leq d2(u, v)$

(C) $d1(u, v) \geq d2(u, v)$

(D) $d1(u, v) \neq d2(u, v)$

2025 CS1 · Q.43 · 2 marks

Let G(V, E) be an undirected and unweighted graph with 100 vertices. Let d(u, v) denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of d(u, v), u, v $\in$ V such that $u \neq v$, be 30. Let T be any breadth- first-search tree of G. Which ONE of the given options is CORRECT for every such graph G?

(A) The height of T is exactly 15.

(B) The height of T is exactly 30.

(C) The height of T is at least 15.

(D) The height of T is at least 30.

2025 CS1 · Q.64 · 2 marks

The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____ . (answer in integer)

2025 CS2 · Q.29 · 1 mark

Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?

(A) A DFS tree of G is a Shortest Path tree of G.

(B) Every non-tree edge of G with respect to a DFS tree is a forward/back edge.

(C) If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.

(D) Both BFS and DFS can be used to find the connected components of G.

2025 CS2 · Q.37 · 2 marks
Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?
(A) Every MST remains an MST, and every SP remains an SP.
(B) MSTs need not remain MSTs, and every SP remains an SP.
(C) Every MST remains an MST, and SPs need not remain SPs.
(D) MSTs need not remain MSTs, and SPs need not remain SPs.

2025 CS2 · Q.59 · 2 marks
Consider the following algorithm someAlgo that takes an undirected graph G as input. someAlgo(G) 1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS. 2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS. 3. Output the distance between u and z in G. The output of someAlgo(T) for the tree shown in the given figure is _____. (Answer in integer)

2024 CS1 · Q.45 · 2 marks
Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v. Suppose T is also a breadth first search (BFS) tree in G, rooted at v. Which of the following statements is/are TRUE for every such graph G and tree T ?
(A) There are no back-edges in G with respect to the tree T
(B) There are no cross-edges in G with respect to the tree T
(C) There are no forward-edges in G with respect to the tree T
(D) The only edges in G are the edges in T

2024 CS1 · Q.60 · 2 marks
The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

2024 CS2 · Q.51 · 2 marks
Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has even weight. Which of the following statements is/are TRUE for every such graph G ?
(A) All edges in G have even weight
(B) All edges in G have even weight OR all edges in G have odd weight
(C) In each cycle C in G, all edges in C have even weight In each cycle C in G, either all edges in C have even weight OR all edges in C
(D) have odd weight

2024 CS2 · Q.59 · 2 marks
The number of distinct minimum-weight spanning trees of the following graph is _____
3 b c 1 2 2 1 2 2 a g d 2 2 1 1 f e 3

2022 · Q.49 · 2 marks
Consider a simple undirected weighted graph G , all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?
(A) The edge with the second smallest weight is always part of any minimum spanning tree of G .
(B) One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G .
(C) Suppose S V be such that S and S V . Consider the edge with the minimum weight such that one of its vertices is in S and the other in V \ S . Such an edge will always be part of any minimum spanning tree of G .
(D) G can have multiple minimum spanning trees.

2022 · Q.58 · 2 marks
Let G (V , E) be a directed graph, where V {1, 2,3, 4,5} is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A. 1, 1≤j≤i≤5 A[i][j] = 0, otherwise A[i][j] = 1 indicates a directed edge from node i to node j. A directed spanning tree of G , rooted at r V , is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V . The number of such directed spanning trees rooted at vertex 5 is_____.

2021 · Q.17 · 1 mark [OCR]
Consider the following undirected graph with edge weights as shown: 09 0.1 0.9 0.1 The number of minimum-weight spanning trees of the graph is

2021 · Q.36 · 2 marks [OCR]
Let G = (V, E) be an undirected unweighted connected graph. The diameter of G is defined as: diam(G) = max {the length of shortest path between u and v} u,ve Let M be the adjacency matrix of G. Define graph G2 on the same set of vertices with adjacency matrix N, where Ni = 1 if M,; >0or P,; > 0, where P = M? 0 otherwise Which one of the following statements is true?
(A) | diam(G,) < [diam(G)/2]
(B) | [diam(G)/2] < diam(G2) < diam(G)
(C) | diam(G.) = diam(G)
(D) | diam(G) < diam(G,) < 2 diam(G) CS, Set-1 - Copyright © GATE 2021

2020 · Q.31 · 2 marks [OCR]
Let G = (V,E) be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighted edge (u,v) € V x V is added to G. The worst case time complexity of determining if T is still an MST of the resultant graph is @) O(|E| + IVI) ®) O(EIIVI) (©) O([E| log |V |)) O(IV|)

2020 · Q.40 · 2 marks [OCR]
Let G=(V,E)be a directed, weighted graph with weight functionw: ER. For some function f:V +R, for each edge (u,v)¢£,define w'(u,v) as w(u,v)+ f(u)— f(r). Which one of the options completes the following sentence so that it is TRUE? “The shortest paths in G under w are shortest paths under w' too,
(A) for every f:V > R (8) ifand only if VueV, f(u) is positive () ifand only if VweV, f(u) is negative (0) if and only if f(w) is the distance from s to wu in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G

2020 · Q.49 · 2 marks [OCR]
Consider a graph G = (V,E), where V = {14, V2,..., Vigo}, EB = {(v.¥)|1 si< j < 100}, and weight of the edge (vj, vj) is |i—j|. The weight of minimum spanning tree of G is

2019 · Q.38 · 2 marks [OCR]
Let G be any connected, weighted, undirected graph. I. G has a unique minimum spanning tree, if no two edges of G have the same weight. IL. G has a unique minimum spanning tree, if, for every cut of G, there is a unique minimum-weight edge crossing the cut. Which of the above two statements is/are TRUE?
(A) Ionly
(B) I only
(C) Both I and I
(D) Neither I nor IL cs 9/15 GATE 2019 Computer Science and Information Technology Set-2

2018 · Q.30 · 2 marks
Let G be a simple undirected graph. Let TD be a depth first search tree of G. Let TB be a breadth first search tree of G. Consider the following statements. (I) No edge of G is a cross edge with respect to TD. (A cross edge in G is between two nodes neither of which is an ancestor of the other in TD.) (II) For every edge (u,v) of G, if u is at depth i and v is at depth j in TB, then |i − j| = 1. Which of the statements above must necessarily be true?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II CS 9/20 GATE 2018

2018 · Q.47 · 2 marks
Consider the following undirected graph G: 4 x 1 3 4 5 4 Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G. The number of MWSTs of G for this value of x is _____.

2017 CS1 · Q.26 · 2 marks [OCR]
Let G = (V, E) be anv connected undirected edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements: (I) Minimum Spanning Tree of G is always unique. (II) Shortest path between any two vertices of G is always unique. Which of the above statements is/are necessarily true?
(A) (I) only
(B) CI) only
(C) both (1) and (ID)
(D) neither (I) nor (I)

2017 CS2 · Q.15 · 1 mark [OCR]
The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?)-—_)—©@) @®) @—®
(A) MNOPQR
(B) NQMPOR
(C) QMNROP
(D) POQNMR =====PAGE 7=====

2016 · Q.11 · 1 mark
Consider the following directed graph: b c a f d e The number of different topological orderings of the vertices of the graph is .

2016 · Q.14 · 1 mark
Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE? P: Minimum spanning tree of G does not change Q: Shortest path between any pair of vertices does not change
(A) P only
(B) Q only
(C) Neither P nor Q
(D) Both P and Q

2016 · Q.38 · 2 marks
Consider the weighted undirected graph with 4 vertices, where the weight of edge {i, j} is given by the entry Wi j in the matrix W . ⌈ ⌉ 0 2 8 5 ⌈ 2 0 5 8 ⌈ W =⌈⌊ 8 5 0 x ⌋ ⌈ 5 8 x 0 The largest possible integer value of x, for which at least one shortest path between some pair of vertices will contain the edge with weight x is .

2016 · Q.39 · 2 marks
Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is .

2016 · Q.40 · 2 marks
G = (V, E) is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE? I. If e is the lightest edge of some cycle in G, then every MST of G includes e II. If e is the heaviest edge of some cycle in G, then every MST of G excludes e
(A) I only
(B) II only
(C) both I and II
(D) neither I nor II

B. New practice questions with solutions

1. (1 mark · NAT) K_4 is a complete graph on 4 vertices with all 6 edges of equal weight. The number of distinct minimum spanning trees is _____.

16. Every spanning tree has the same weight, so all $4^2 = 16$ labelled spanning trees are

minimum.

2. (2 marks · MSQ) G is a connected undirected graph with distinct positive edge weights. Every weight w is replaced by w^2. Which statements are TRUE?

A. The minimum spanning tree stays the same.
B. The shortest-path tree from any source stays the same.
C. The shortest-path distance between any two vertices becomes the square of its old value.
D. The ordering of the edges by weight is unchanged.

(A), (D). Squaring is increasing on positive numbers, so the edge order and the MST are preserved. Path sums do not behave like that: take s–b with weight 3 and s–a–b with weights 2 and 2. Before squaring the direct edge wins ($3 < 4$). After squaring the two-edge path wins ($4 + 4 = 8 < 9$). So (B) and (C) are false.

Asymptotics & recurrences

Order-of-growth ordering, solving recurrences. · 17 questions, 23 marks over 15 papers (≈1.5 marks/paper)

A. Past-paper questions (17)

2026 CS1 · Q.17 · 1 mark
Consider the following recurrence relations: For all n > 1, n T1 (n) = 4T1 ( ) + T2 (n) 2 n T2 (n) = 5T2 ( ) + Θ(log2 n) 4 Assume that for all n ≤ 1, T1 (n) = 1 and T2 (n) = 1. Which one of the following options is correct?
(A) T1 (n) = Θ(n2)
(B) T1 (n) = Θ(n2 log2 n)
(C) T1 (n) = Θ(nlog4 5)
(D) T1 (n) = Θ(nlog4 5 log2 n)

2026 CS2 · Q.24 · 1 mark
Consider the following functions, where n is a positive integer. n1/3 , log(n) , log(n!), 2log(n) Which one of the following options lists the functions in increasing order of asymptotic growth rate? Note: Assume the base of log to be 2.
(A) log(n) , n1/3 , 2log(n) , log(n!)
(B) n1/3 , log(n) , log(n!) , 2log(n)
(C) log(n) , n1/3 , log(n!) , 2log(n)
(D) 2log(n) , n1/3 , log(n) , log(n!)

2026 CS2 · Q.25 · 1 mark
Which of the following can be recurrence relation(s) corresponding to an algorithm with time complexity Θ(n)?
(A) T(n) = T(n − 1) + 1, T(1) = 1
(B) n T(n) = 2T ( ) + 1, T(1) = 1 2
(C) n T(n) = 2T ( ) + n, T(1) = 1 2
(D) T(n) = T(n − 1) + n, T(1) = 1

2026 CS2 · Q.38 · 2 marks
Consider an array A of integers of size n. The indices of A run from 1 to n. An algorithm is to be designed to check whether A satisfies the condition given below. ∀i, j ∈ {1, ... , n − 1} such that i > j, (A[i + 1] − A[i]) > (A[j + 1] − A[j]) Which one of the following gives the worst case time complexity of the fastest algorithm that can be designed for the problem?
(A) Θ(n)
(B) Θ(log(n))
(C) Θ(n log(n))
(D) Θ(n2)

2025 CS1 · Q.20 · 1 mark
T(n) = 2T(n − 1) + n2n for n > 0, T(0) = 1. Which ONE of the following options is CORRECT?
(A) T(n) = Θ(n2 2n)
(B) T(n) = Θ(n2n)
(C) T(n) = Θ((log n)2 2n)
(D) T(n) = Θ(4n)

2024 CS1 · Q.17 · 1 mark

Given an integer array of size N, we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

(A) both O(N) and Ω(N)
(B) O(N) but not Ω(N)
(C) Ω(N) but not O(N)
(D) neither O(N) nor Ω(N)

2024 CS1 · Q.42 · 2 marks

Consider the following recurrence relation: T(n) = {√nT(√n) + n for n ≥ 1, 1 for n = 1. Which one of the following options is CORRECT?

(A) T(n) = Θ(n log log n)
(B) T(n) = Θ(n log n)
(C) T(n) = Θ(n2 log n)
(D) T(n) = Θ(n2 log log n)

2023 · Q.29 · 1 mark

Let f and g be functions of natural numbers given by f (n) = n and g(n) = n2 . Which of the following statements is/are TRUE?

(A) f ∈ O(g)
(B) f ∈ Ω(g)
(C) f ∈ o(g)
(D) f ∈ Θ(g) CS

2023 · Q.54 · 2 marks

Consider functions Function 1 and Function 2 expressed in pseudocode as follows: Function 1 Function 2 while n > 1 do for i = 1 to 100 ∗ n do for i = 1 to n do x = x + 1; x = x + 1; end for end for n = n/2; end while Let f1 (n) and f2 (n) denote the number of times the statement “x = x + 1” is executed in Function 1 and Function 2, respectively. Which of the following statements is/are TRUE?

(A) f1 (n) ∈ Θ(f2 (n))
(B) f1 (n) ∈ o(f2 (n))
(C) f1 (n) ∈ ω(f2 (n))
(D) f1 (n) ∈ O(n) CS

2022 · Q.11 · 1 mark

Which one of the following statements is TRUE for all positive functions f (n) ?

(A) f (n2) (f (n)2), when f (n) is a polynomial
(B) f (n2) o(f (n)2)
(C) f (n2) O(f (n)2), when f (n) is an exponential function
(D) f (n2) (f (n)2)

2022 · Q.51 · 2 marks
Consider the following recurrence: f (1) 1; f (2n) 2 f (n) 1, for n 1; f (2n 1) 2 f (n) 1, for n 1. Then, which of the following statements is/are TRUE?
(A) f (2n 1) 2n 1
(B) f (2n) 1
(C) f (5 2n) 2n1 1
(D) f (2n 1) 2n 1

2021 · Q.3 · 1 mark [OCR]
Consider the following three functions. fi=10" fo=ne" fga=nTM Which one of the following options arranges the functions in the increasing order of asymptotic growth rate?
(A) | fs, fa, fi
(B) | fa, fis fs
(C) | fi, fa, fs
(D) | fa. fa. fi QA Consider the following statements. Sy: The sequence of procedure calls corresponds to a preorder traversal of the activation tree. So: The sequence of procedure returns corresponds to a postorder traversal of the activation tree. Which one of the following options is correct?
(A) | S, is true and Sy is false
(B) | Si is false and S2 is true
(C) | S, is true and Sy is true
(D) | Si is false and Sp is false CS, Set-1 - Copyright © GATE 2021

2021 · Q.30 · 2 marks [OCR]
Consider the following recurrence relation. — a +T(2n/5)+7n ifn >0 1 ifn=0 Which one of the following options is correct?
(A) | T(n) = @(n/?)
(B) | T(n) = O(nlogn)
(C) | T(n) = O(n)
(D) | T(n) = O((logn)*/?) CS, Set-1 - Copyright © GATE 2021

2020 · Q.2 · 1 mark [OCR]
For parameters aand b, both of which are @(1), T(n)=T(n'")+1, and T(b)=1. Then 7(n) is
(A) O(log, log, n) 6) O(log, ”) ©) O(log, log, n) (0) O(log, log, n)

2017 CS1 · Q.4 · 1 mark [OCR]
Consider the following functions from positive integers to real numbers: 100 10, vn. n, log,n. —. n The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is: 100
(A) log,n. ——. 10. vn.n i 100
(B) —.10. log,n. vn. n nN 100 © 10, —, vn, log, n,n n
(D) = log, 7. 10. Jn. n n =====PAGE 4=====

2017 CS2 · Q.3 · 1 mark [OCR]
Match the algorithms with their time complexities: Algorithm Time complexity (P) Towers of Hanoi with 7 disks (i) O(n?) (Q) Binary search given 7 sorted numbers (ii) O(n logn) (R) Heap sort given 7 numbers at the worst case (iii) @(2") (S) Addition of two n X n matrices (iv) O(log 72)
(A) P > (iii), Q > (iv). R @. S > Gi)
(B) P > (iv). Q = (iii), R > (i), S > (ii)
(C) P = (iii), Q > (iv), R> (ii). S > i)
(D) P > (iv), Q— (iii), R > (ii), S > (i) =====PAGE 3=====

2017 CS2 · Q.30 · 2 marks [OCR]
Consider the recurrence function T(n) = oo woe 2; O0<n<2 Then T(n) in terms of © notation is
(A) O(log log n)
(B) O(log n)
(C) O(n)
(D) @(n) =====PAGE 12=====

B. New practice questions with solutions

1. (1 mark · MCQ) Which option lists the functions in increasing order of asymptotic growth?

A. $(\log n)^3$, $n \log n$, $n^{1.5}$, $2^{\sqrt{n}}$
B. $n \log n$, $(\log n)^3$, $n^{1.5}$, $2^{\sqrt{n}}$
C. $(\log n)^3$, $n^{1.5}$, $n \log n$, $2^{\sqrt{n}}$
D. $(\log n)^3$, $n \log n$, $2^{\sqrt{n}}$, $n^{1.5}$

(A). Polylogarithms grow slower than $n \log n$, which is slower than $n^{1.5}$. The function $2^{\sqrt{n}}$ beats every polynomial.

2. (2 marks · MCQ) $T(n) = 2T(n/2) + n/\log n$ with $T(1) = 1$. Then $T(n)$ is

A. $\Theta(n)$ (B) $\Theta(n \log \log n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2)$

(B). The Master theorem does not apply. Summing level by level gives $n \cdot \Sigma\, 1/(\log n - i) = n \cdot H(\log n) = \Theta(n \log \log n)$.

Sorting, searching, divide & conquer

Sorting complexity, searching, divide-and-conquer, comparison counts. · 11 questions, 14 marks over 15 papers (≈0.9 marks/paper)

A. Past-paper questions (11)

2026 CS2 · Q.32 · 1 mark

Consider an array A = [10, 7, 8, 19, 41, 35, 25, 31]. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations. A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R. The number of void merge operations among these 7 merge operations is _____. (answer in integer)

2025 CS1 · Q.33 · 1 mark

The pseudocode of a function fun() is given below: fun(int A[0,...,n-1]){ for i=0 to n-2 for j=0 to n-i-2 if (A[j]>A[j+1]) then swap A[j] and A[j+1] } Let A[0, ... ,29] be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function fun() is called with A[0, ... ,29] as argument, is _____. (Answer in integer)

2025 CS2 · Q.20 · 1 mark

Consider an unordered list of N distinct integers. What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?
(A) 1
(B) N−1
(C) N
(D) 2N − 1

2025 CS2 · Q.41 · 2 marks

An array A of length n with distinct elements is said to be bitonic if there is an index 1 ≤ i ≤ n such that A[1. . i] is sorted in the non-decreasing order and A[i + 1 . . n] is sorted in the non-increasing order. Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A?
(A) Θ(n)
(B) Θ(1)
(C) Θ(log 2 n)
(D) Θ(log n)

2024 CS2 · Q.35 · 1 mark

Let Abe an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array [2, 5, 3, 1, 4, 2, 6] is _____

2021 · Q.2 · 1 mark [OCR]
Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary array of n elements. Which one of the following choices is correct?
(A) |}t>2n-2
(B) |t>3 [5] and t < 2n—2
(C) |t>nandt<3 [5]
(D) | t > flog,(n)] andt <n CS, Set-1 - Copyright © GATE 2021

2021 · Q.9 · 1 mark [OCR]
Consider the following array. [23 | 32 | 45 | 69 | 72 | 73 | 89 | 97] Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?
(A) | Selection sort
(B) | Mergesort
(C) | Insertion sort
(D) | Quicksort using the last element as pivot

2019 · Q.20 · 1 mark [OCR]
An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is

2019 · Q.37 · 2 marks [OCR]
There are n unsorted arrays: 41, A2, ..., 4n. Assume that 7 is odd. Each of Ai, 42, ..., An contains 7 distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of Ai, 42, ..., An is
(A) O(n)
(B) O(7 log n)
(C) O(n”) () AQ(n? log n)

2017 CS1 · Q.48 · 2 marks [OCR]
Let A be an array of 31 numbers consisting of a sequence of 0’s followed by a sequence of 1’s. The problem is to find the smallest index 7 such that 4/7] is 1 by probing the minimum number of locations in A. The worst case number of probes performed by an optimal algorithm is =====PAGE 22=====

2016 · Q.13 · 1 mark
The worst case running times of Insertion sort, Merge sort and Quick sort, respectively, are:
(A) Θ(n log n), Θ(n log n), and Θ(n2)
(B) Θ(n2), Θ(n2), and Θ(n log n)
(C) Θ(n2), Θ(n log n), and Θ(n log n)
(D) Θ(n2), Θ(n log n), and Θ(n2)

B. New practice questions with solutions

1. (1 mark · NAT) The minimum number of comparisons needed in the worst case to find

both the largest and the smallest of 100 distinct numbers, using the best known pairing method, is _____.

> **148.** $\lceil 3n/2 \rceil - 2 = 150 - 2$.

2. (2 marks · NAT) Top-down merge sort is applied to 8 distinct elements. The maximum number of element comparisons is _____.

> **17.** $C(2) = 1$, $C(4) = 2 \cdot 1 + 3 = 5$, $C(8) = 2 \cdot 5 + 7 = 17$.

Hashing

Open addressing / double hashing, collisions. · 6 questions, 7 marks over 15 papers (≈0.5 marks/paper)

A. Past-paper questions (6)

2026 CS1 · Q.24 · 1 mark

Consider a hash table P[0, 1, ... , 10] that is initially empty. The hash table is maintained using open addressing with linear probing. The hash function used is h(x) = (x + 7) mod 11. Consider the following sequence of insertions performed on P: 1, 13, 22, 15, 11, 24 Which of the following positions in the hash table is/are empty after these insertions are performed?
(A) 0
(B) 10
(C) 2
(D) 1

2026 CS2 · Q.30 · 1 mark

The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function h(k) = k mod 9. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____. (answer in integer)

2025 CS1 · Q.65 · 2 marks

In a double hashing scheme, h1 (k) = k mod 11 and h2 (k) = 1 + (k mod 7) are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is [h1 (k) + i h2 (k)] mod m. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (Answer in integer)

2023 · Q.20 · 1 mark

An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and k > m. Which one of the following is the best hashing strategy to counteract the adversary?
(A) Division method, i.e., use the hash function h(k) = k mod m.
(B) Multiplication method, i.e., use the hash function h(k) = m(kA − kA), where A is a carefully chosen constant.
(C) Universal hashing method.
(D) If k is a prime number, use Division method. Otherwise, use Multiplication method. CS

2022 · Q.16 · 1 mark

Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h1 and h2 . Further suppose our hashing scheme uses h1 for the odd keys and h2 for the even keys. What is the expected number of keys in a slot?
(A) m n
(B) n m
(C) 2n m
(D) n 2m

> **2020 · Q.23 · 1 mark** [OCR]
> Consider a double hashing scheme in which the primary hash function is h,(k) =k mod 23, and the secondary hash function is h,(k)=1+(k mod 19). Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value k =90 is

B. New practice questions with solutions

1. (1 mark · NAT) A table of size 7 uses h(k) = k mod 7 with linear probing. The keys 15, 22, 8, 29 are inserted in that order. The index where 29 is stored is _____.

> **4.** All four keys hash to index 1, so they fill indices 1, 2, 3 and 4 in order.

2. (2 marks · NAT) A table of size 11 uses double hashing with $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 10)$. The keys 25, 36, 47 are inserted in that order. The index where 47 is stored is _____.

> **0.** All three keys have $h_1 = 3$. Key 25 goes to index 3. Key 36 has $h_2 = 7$, so it tries 3, then 10 and is stored at 10. Key 47 has $h_2 = 8$, so it tries 3, then $(3 + 8) \bmod 11 = 0$ and is stored at 0.

Dynamic programming

Dynamic-programming tables, subsequence problems. · 3 questions, 5 marks over 15 papers (≈0.3 marks/paper)

A. Past-paper questions (3)

2026 CS2 · Q.39 · 2 marks
Consider a table T, where the elements T[i][j], 0 ≤ i, j ≤ n, represent the cost of the optimal solutions of different subproblems of a problem that is being solved using a dynamic programming algorithm. The recursive formulation to compute the table entries is as follows: T[0][k] = T[k][0] = 1 for k = 0,1,2, ... , n T[i][j] = 2T[i − 1][j] + 3T[i][j − 1] for 1 ≤ i, j ≤ n Consider the following two algorithms to compute entries of T. Assume that for both the algorithms, for all 0 ≤ i, j ≤ n, T[i][j] has been initialized to 1. Algorithm B1: For i = 1, 2, ... , n For j = 1, 2, ... , n T[i][j] = 2T[i − 1][j] + 3T[i][j − 1] Algorithm B2: For s = 2, 3, ... , 2n For i = 1, 2, ... , n For j = 1, 2, ... , n If (i + j == s) T[i][j] = 2T[i − 1][j] + 3T[i][j − 1] Algorithm Bk , k ∈ {1,2} is said to be correct if and only if it calculates the correct values of T[i][j], for all 0 ≤ i, j ≤ n, (as per the recursive formulation) at the end of the execution of the algorithm Bk . Which one of the following statements is true?
(A) Both algorithms B1 and B2 are correct
(B) Algorithm B1 is correct, but algorithm B2 is incorrect
(C) Algorithm B2 is correct, but algorithm B1 is incorrect
(D) Both algorithms B1 and B2 are incorrect

2019 · Q.25 · 1 mark [OCR]
Consider a sequence of 14 elements: A = [—5S, —10, 6, 3, -1, —2, 13, 4, -9, -1, 4, 12, -3, 0]. The subsequence sum S(i,j) = Y}_;A[k] . Determine the maximum of S(i,/), where 0 <i <j < 14. (Divide and conquer approach may be used.) Answer:

2018 · Q.31 · 2 marks
Assume that multiplying a matrix G1 of dimension p × q with another matrix G2 of dimension q × r requires pqr scalar multiplications. Computing the product of n matrices G1G2G3...Gn can be done by parenthesizing in different ways. Define Gi Gi+1 as an explicitly computed pair for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain G1G2G3G4G5G6 using parenthesization (G1(G2G3))(G4(G5G6)), G2G3 and G5G6 are the only explicitly computed pairs. Consider a matrix multiplication chain F1F2F3F4F5, where matrices F1, F2, F3, F4 and F5 are of dimensions 2×25, 25×3, 3×16, 16×1 and 1×1000, respectively. In the parenthesization of F1F2F3F4F5 that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are
(A) F1F2 and F3F4 only
(B) F2F3 only
(C) F3F4 only
(D) F1F2 and F4F5 only

B. New practice questions with solutions

1. (1 mark · NAT) The length of the longest common subsequence of PQRSTQP and QPSRQTP is _____.

> **4.** One example is Q, S, T, P.

2. (2 marks · NAT) A person climbs a staircase of 10 steps taking 1, 2 or 3 steps at a time. The number of distinct ways to reach the top is _____.

> **274.** Let a(n) = a(n−1) + a(n−2) + a(n−3) with a(0) = 1. The sequence is 1, 1, 2, 4, 7, 13, 24, 44, 81, 149, 274.

3. (2 marks · NAT) A 0/1 knapsack has capacity 9 and four items with (weight, value) = (3, 4), (4, 5), (5, 7), (2, 3). The maximum total value is _____.

> **12.** Items {3, 4, 2} give 4 + 5 + 3 = 12 and items {4, 5} give 5 + 7 = 12. No combination with weight at most 9 does better.

Greedy

Greedy method, Huffman, knapsack variants. · 3 questions, 5 marks over 15 papers (≈0.3 marks/paper)

A. Past-paper questions (3)

> **2018 · Q.48 · 2 marks**
> Consider the weights and values of items listed below. Note that there is only one unit of each item. Item number Weight Value (in Kgs) (in Rupees) 1 10 60 2 7 28 3 4 20 4 2 24 The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by Vopt . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by Vgreedy . The value of Vopt − Vgreedy is _____.

> **2017 CS1 · Q.5 · 1 mark** [OCR]
> Consider the following table: Algorithms Design Paradigms (P) Kruskal (i) Divide and Conquer (Q) Quicksort (ii) Greedy (R) Floyd-Warshall (iii) Dynamic Programming Match the algorithms to the design paradigms they are based on. (A)®) - G@). (Q) < Gi). R) o @ B) P)< Gi). Qo @. (QR) (i) ©) Mo@. Qe@. ®¢ tii) 0) Me@ Qe). BR) o Gi)

> **2017 CS2 · Q.50 · 2 marks** [OCR]
> A message is made up entirely of characters from the set X = {P,Q,R,S,T}. The table of probabilities for each of the characters is shown below: Character Probability P 0.22 Q 0.34 R 0.17 S 0.19 T 0.08 Total 1.00 If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is =====PAGE 21=====

B. New practice questions with solutions

1. (1 mark · NAT) Huffman coding is applied to symbols with frequencies 5, 9, 12, 13, 16, 45. The weighted code length Σ(frequency × code length) is _____.

> **224.** The merges cost 14, 25, 30, 55 and 100, and 14 + 25 + 30 + 55 + 100 = 224.

2. (2 marks · NAT) Five unit-time jobs have (profit, deadline) = (100, 2), (19, 1), (27, 2), (25, 1), (15, 3). The maximum total profit from job sequencing with deadlines is _____.

> **142.** Schedule the 27-profit job in slot 1, the 100-profit job in slot 2 and the 15-profit job in slot 3. The two deadline-1 jobs cannot also fit. 100 + 27 + 15 = 142.